

Bridging the Gap Between Theory and Practice in Off-Policy Evaluation

Open-Source Dataset/Software and Application in Fashion E-Commerce

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Outline

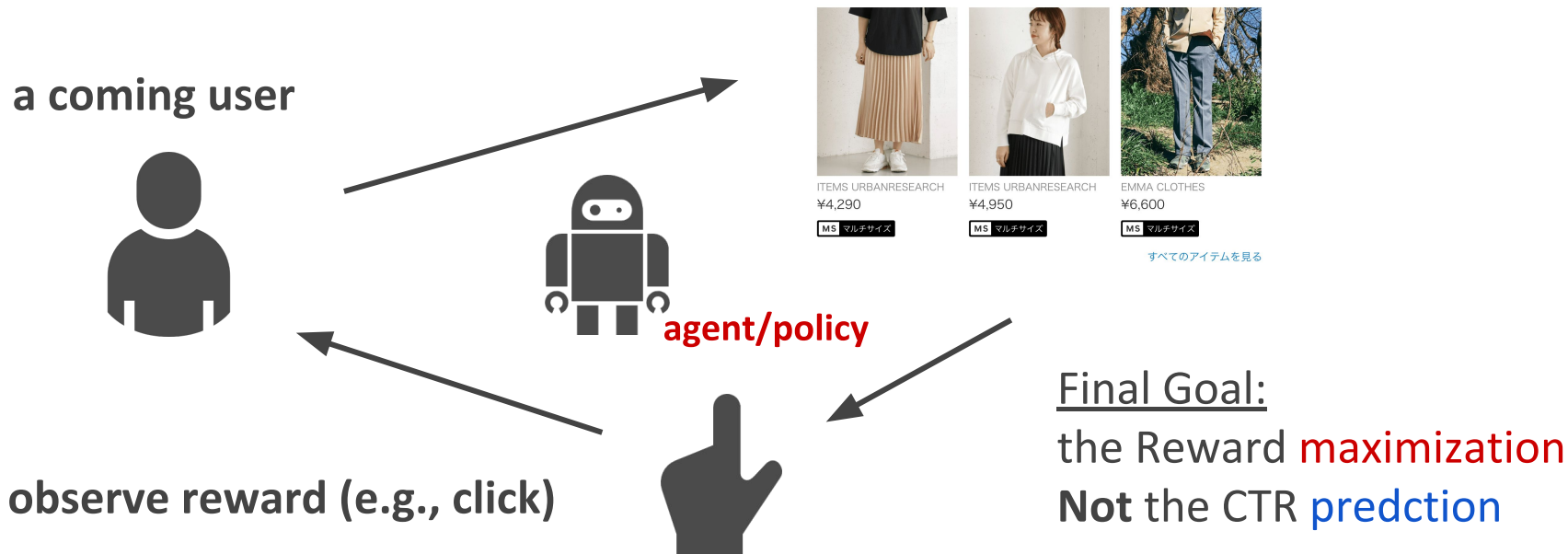
- Off-Policy Evaluation (OPE)
 - Basics and Current Issues
- Open Bandit Project
 - Open Source Dataset and Software for OPE
 - A Live Demonstration about OPE with the Open-Source
- Applications on the Large Fashion E-Commerce Platform
 - Off-Policy Estimator Selection / Counterfactual Policy Search

Off-Policy Evaluation: Basics and Issues

Machine Learning for Decision Making (Bandit / RL)

We often use machine learning to make **decisions**, not predictions

Decision Making: item recommendation



Many Applications of “Machine Decision Making”

- news recommendation (by Yahoo)
- music/playlist/podcast recommendation (by Spotify)
- artwork personalization (by Netflix)
- ad allocation optimization (by Criteo)

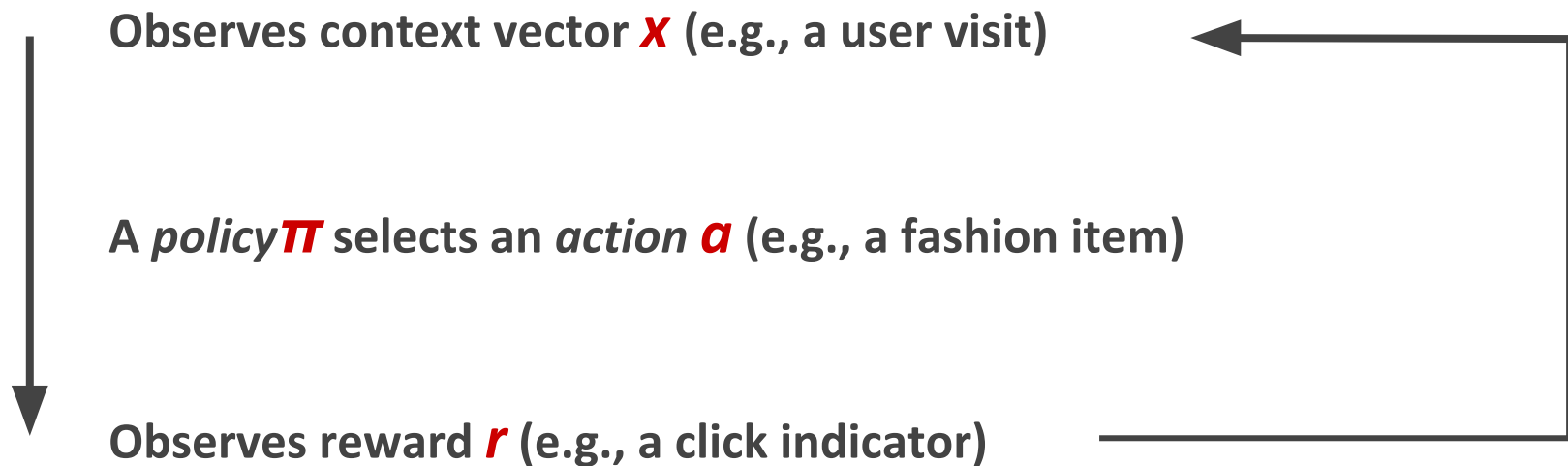
Motivation of OPE

- medicine
- education

etc...

We want to evaluate the performance of
a *new decision making policy* using data
generated by a *behavior, past policy*

Data Generating Process (contextual bandit setting)



a *policy* interacts with the environment
and produces the *log data* valuable for *redesigning the system*

Logged Bandit Feedback

a behavior (or past) policy gives us *logged bandit feedback*

$$\mathcal{D} := \left\{ (x_i, a_i, r_i) \right\}_{i=1}^n$$

$$a_i \sim \pi_b (a \mid x_i)$$

action choice by behavior policy

$$r_i \sim p (r \mid x_i, a_i)$$

observed reward

Estimation Target in Off-Policy Evaluation (OPE)

In OPE, we want to estimate the *policy value (policy performance)* of an *evaluation (or new) policy*

$$\underbrace{V(\pi_e)}_{\text{policy value}} := \mathbb{E}_{p(x) \underbrace{\pi_e(a|x)}_{\text{evaluation policy}} p(r|x,a)} [r]$$

➡ **expected reward obtained by running π_e on a real system**

e.g., **expected revenue** by (hypothetically) deploying a new policy

Benefits of Off-Policy Evaluation

Goal: Accurate OPE of the policy value of an evaluation policy

$$V(\pi_e) \approx \hat{V}(\pi_e; \mathcal{D})$$
$$\mathcal{D} \sim p(x)\pi_b(a|x)p(r|x, a)$$

- avoid deploying poor performing policies without A/B tests
- identify promising new policies among many candidates

etc...

Growing interest in OPE!

Direct Method (DM)

DM first estimates the expected reward and uses it to estimate the policy value

$$\hat{V}_{\text{DM}}(\pi_e; \mathcal{D}, \hat{q}) := \mathbb{E}_n \left[\sum_{a \in \mathcal{A}} \pi_e(a \mid x_i) \underbrace{\hat{q}(x_i, a)}_{\text{reward estimator}} \right]$$

- **Large bias** when the model is mis-specified
- **Small variance**

$$\mathbb{E}[r \mid x, a] \approx \hat{q}(x, a)$$

$$\mathbb{E}_n[\cdot] : \text{empirical expectation over } \mathcal{D}$$

Inverse Probability Weighting (IPW) Estimator

IPW re-weights observed rewards by importance weights

$$\hat{V}_{\text{IPW}}(\pi_e; \mathcal{D}) := \mathbb{E}_n \left[\underbrace{w(x_i, a_i)}_{\text{importance weight}} r_i \right]$$

importance weight

$$w(x, a) := \pi_e(a \mid x) / \pi_b(a \mid x)$$

- **Consistent**
- **Large variance** when old and new policies are largely different

Doubly Robust (DR) Estimator

DR uses DM as a baseline and applies IPW to shifted rewards

$$\hat{V}_{\text{DR}}(\pi_e; \mathcal{D}, \hat{q}) \\ := \underbrace{\hat{V}_{\text{DM}}(\pi_e; \mathcal{D}, \hat{q})}_{\text{baseline}} + \mathbb{E}_n \left[\underbrace{w(x_i, a_i) (r_i - \hat{q}(x_i, a_i))}_{\text{weighted shifted reward}} \right]$$

$$\mathbb{E}[r \mid x, a] \approx \hat{q}(x, a)$$

- **Consistent**
- **Locally Efficient**
(desirable variance)

Theoretical/Methodological Advances in OPE

- Self-Normalized IPW [[Swaminathan and Joachims 2015](#)]
- Switch Doubly Robust Estimator [[Wang+ 2017](#)]
- More Robust Doubly Robust Estimator [[Farajtabar+ 2018](#)]
- Hirano-Imbens-Ridder Estimator [[Narita+ 2019](#)]
- Continuous Adaptive Blending [[Su+ 2019](#)]
- REG and EMP [[Kallus & Uehara 2019](#)]
- Doubly Robust with Shrinkage [[Su+ 2020](#)]

**It seems the OPE community
have made great progress
over the years!**

There are many other estimators in the reinforcement learning setting

Theoretical/Methodological Advances in OPE

- Self-Normalized IPW [[Swaminathan and Joachims 2015](#)]
- Switch Doubly Robust Estimator [[Wang+ 2017](#)]
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**Does this trend really
expect the “era of OPE”?**

There are many other estimators in the reinforcement learning setting

Issues with the current experimental procedures...

Experiments in *every* OPE paper rely on either

1. Synthetic or classification data (unrealistic)

or

2. (Real, but) Unpublished data (irreproducible)

there is the *critical gap between theory and practice in OPE*

Open Bandit Project:
Public Dataset and Software

The Goal of the “Open Bandit Project”

We enable *realistic* and *reproducible* experiments on

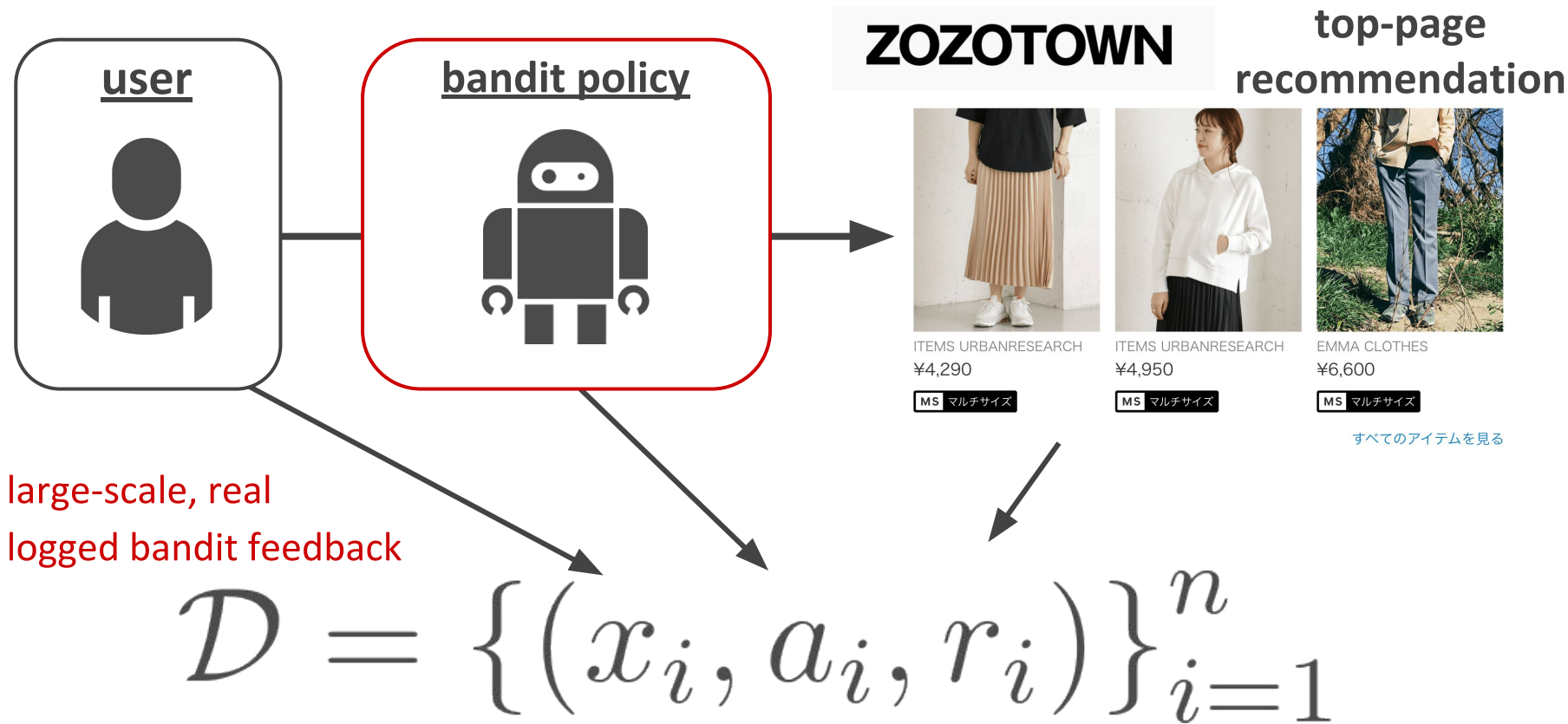
- **Off-Policy Evaluation (OPE)**
- (Bandit Algorithms)



“Open Bandit Dataset”

and ***“Open Bandit Pipeline”***

Open Bandit Dataset: Data Collection on ZOZOTOWN



Open Bandit Dataset: Release Page

Open Bandit Dataset

Open Bandit Dataset is a public real-world logged bandit feedback data. The dataset is provided by ZOZO, Inc., the largest Japanese fashion e-commerce company with over 5 billion USD market capitalization (as of May 2020). The company uses multi-armed bandit algorithms to recommend fashion items to users in a large-scale fashion e-commerce platform called ZOZOTOWN.

This dataset is released along with the paper:

Yuta Saito, Shunsuke Aihara, Megumi Matsutani, Yusuke Narita.

Large-scale Open Dataset, Pipeline, and Benchmark for Bandit Algorithms <https://arxiv.org/abs/2008.07146>

When using this dataset, please cite the paper with following bibtex:

```
@article{saito2020large,
  title={Large-scale Open Dataset, Pipeline, and Benchmark for Bandit Algorithms},
  author={Saito, Yuta, Shunsuke Aihara, Megumi Matsutani, Yusuke Narita},
  journal={arXiv preprint arXiv:2008.07146},
  year={2020}
}
```

Data description

Open Bandit Dataset is constructed in an A/B test of two multi-armed bandit policies in a large-scale fashion e-commerce platform, ZOZOTOWN. It currently consists of a total of 26M rows, each one representing a user impression with some feature values, selected items as actions, true propensity scores, and click indicators as an outcome. This is especially suitable for evaluating *off-policy evaluation* (OPE), which attempts to estimate the counterfactual performance of hypothetical algorithms using data generated by a different algorithm in use.

<https://research.zozo.com/data.html>

Fields

Here is a detailed description of the fields (they are comma-separated in the CSV files):

{behavior_policy}/{campaign}.csv (behavior_policy in (bts, random), campaign in (all, men, women))

- timestamp: timestamps of impressions.
- item_id: index of items as arms (index ranges from 0-80 in "All" campaign, 0-33 for "Men" campaign, and 0-46 "Women" campaign).
- position: the position of an item being recommended (1, 2, or 3 correspond to left, center, and right position of the ZOZOTOWN recommendation interface, respectively).
- click: target variable that indicates if an item was clicked (1) or not (0).
- propensity_score: the probability of an item being recommended at each position.
- user feature 0-4: user-related feature values.
- user-item affinity 0-: user-item affinity scores induced by the number of past clicks observed between each user-item pair.

item_context.csv

- item_id: index of items as arms (index ranges from 0-80 in "All" campaign, 0-33 for "Men" campaign, and 0-46 "Women" campaign).
- item feature 0-3: item related feature values

Please visit the [examples](#) to learn how to use the data.

Google Group

Open Bandit Dataset: Schema Image

	a		π_b	r	x	
timestamp	item_id	position	action prob	click indicator	features	...
2019-11-xx	25	1	0.0002	0	e2500f3f	...
2019-11-xx	32	2	0.043	1	7c414ef7	...
2019-11-xx	11	3	0.167	0	60bd4df9	...
2019-11-xx	40	1	0.0011	0	7c20d9b5	...
...	...	...	...	...	...	...

Open Bandit Dataset: Essential Features

- **over 25M records** collected by online experiments of bandit algorithms on a large-scale fashion e-commerce (*ZOZOTOWN*)
- **two sets of logged bandit feedback** collected by running **multiple different bandit policies** **most important**
 - *Uniform Random/Thompson Sampling* (fixed)



realistic and reproducible **“evaluation of OPE”**

for the first time

Open Bandit Dataset: Protocol for the “Evaluation of OPE”

We can use our Open Bandit Dataset for the “evaluation of OPE”

evaluate the estimation accuracy of
an OPE estimator in a data-driven manner

1. Prepare *two* sets of logged bandit feedback
2. Evaluate the performance of an eval policy by an OPE estimator
3. Calculate the ground-truth performance of the eval policy by the *on-policy estimation*
4. Compare the ground-truth with the value estimated by OPE to evaluate the estimation accuracy of the estimator

Protocol for the Evaluation of OPE with Open Bandit Dataset

1. Our data have logged bandit feedback collected by *two different policies*

$$\mathcal{D}_e \sim p(x) \underbrace{\pi_e(a \mid x)}_{\text{e.g.) thompson sampling}} p(r \mid x, a)$$

e.g.) thompson sampling

$$\mathcal{D}_b \sim p(x) \underbrace{\pi_b(a \mid x)}_{\text{e.g.) uniform random}} p(r \mid x, a)$$

e.g.) uniform random

Protocol for the Evaluation of OPE with Open Bandit Dataset

2. Regard one policy as an **evaluation policy** and the other as a **behavior policy**. Then, estimate the performance of the evaluation policy by OPE

$$V(\pi_e) \approx \underline{\hat{V}}(\pi_e; \mathcal{D}_b)$$

e.g.) DM/IPW/DR

π_e : **evaluation policy**

π_b : **behavior policy**

- The task here is to evaluate the estimation accuracy of $\hat{V}$

Protocol for the Evaluation of OPE with Open Bandit Dataset

3. Regard the **on-policy estimation** of the policy value of the evaluation policy as the ground-truth policy value

$$V_{\text{on}}(\pi_e; \underline{\mathcal{D}_e}) := \mathbb{E}_{n_e}[r_i]$$

collected by running the **evaluation policy**

we can do this on-policy estimation, as we have $\mathcal{D}_e$ in our dataset

Protocol for the Evaluation of OPE with Open Bandit Dataset

4. Compare the estimated policy value with the ground-truth to evaluate the OPE estimator, for example, using the *squared error*

squared error of $\hat{V}$ (performance metric of an OPE estimator; lower = accurate)

$$:= \left(\underbrace{V_{\text{on}}(\pi_e; \mathcal{D}_e)}_{\text{ground-truth policy value}} - \underbrace{\hat{V}(\pi_e; \mathcal{D}_b)}_{\text{value estimated by OPE}} \right)^2$$

By applying this procedure to several estimators,
we can evaluate and compare different OPE methods

Comparison with Existing Real-World Bandit Datasets

Table 2: Comparison of Currently Available Large-scale Bandit Datasets

	Criteo Data (Lefortier et al. 2016)	Yahoo! Data (Li et al. 2010)	Open Bandit Dataset (ours)
Domain	Display Advertising	News Recommendation	Fashion E-Commerce
#Data	$\geq 103\text{M}$	$\geq 40\text{M}$	$\geq 26\text{M}$ (will increase)
#Behavior Policies	1	1	2 (will increase)
Random A/B Test Data	✗	✓	✓
Behavior Policy Code	✗	✗	✓
Evaluation of Bandit Algorithms	✓	✓	✓
Evaluation of OPE	✗	✗	✓
Pipeline Implementation	✗	✗	✓

Our Open Bandit Dataset

- contains **multiple** different policies
- enables **the evaluation of OPE** for the first time
- comes with the pipeline implementations (Open Bandit Pipeline)

**the existing datasets cannot
compare different OPE estimators**

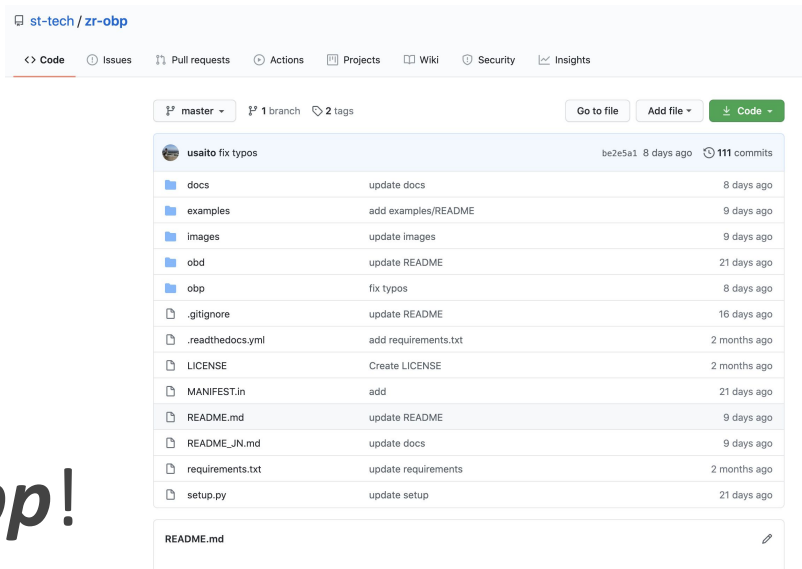
Open Bandit Pipeline (OBP)

We have implemented *Open Bandit Pipeline (OBP)*
to streamline and standardize experiments on OPE



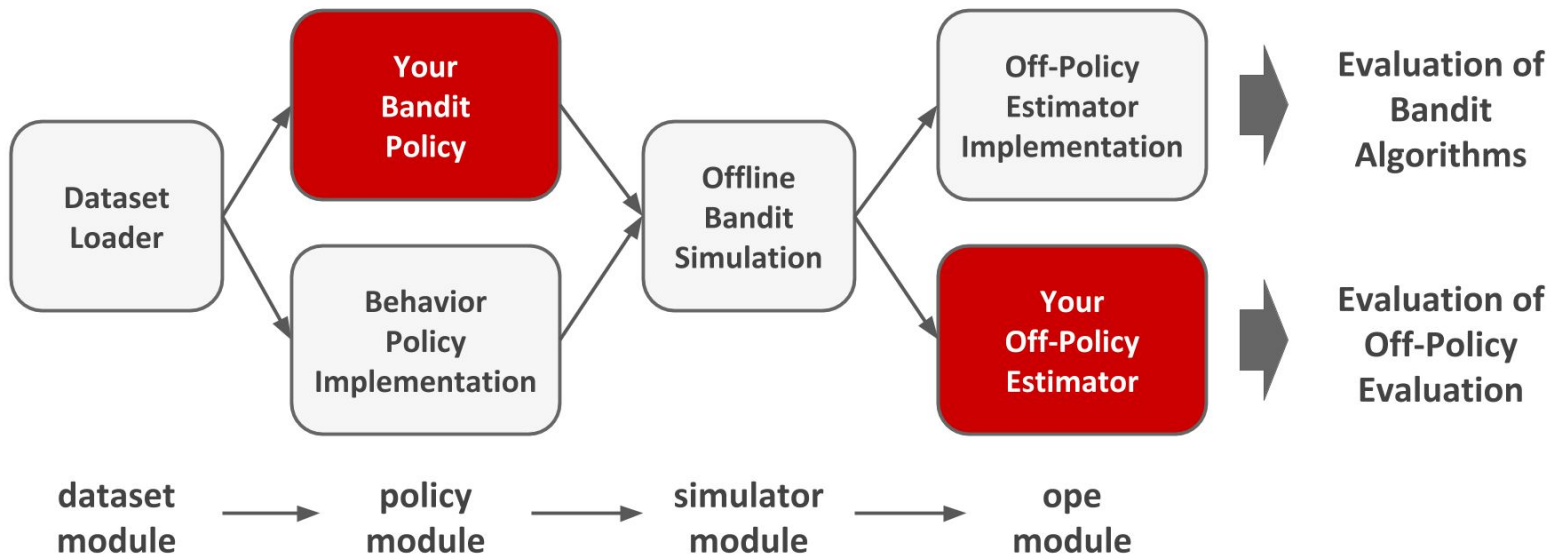
**OPEN
BANDIT
PIPELINE™**

find out *zr-obp*!



Open Bandit Pipeline: Main Modules

OBP consists of **four main modules** (dataset, policy, simulator, and ope)



Proof of Concept Analysis with Our Data and Pipeline

Live OPE demonstration with  **OPEN BANDIT PIPELINE™** (our pipeline software)

1. Load and Preprocess **Open Bandit Dataset** (dataset module)
2. Train a **counterfactual policy** based on the logged bandit feedback (policy module)
3. **Off-Policy Evaluation** of the counterfactual policy (ope module)

Q. Should ZOZOTOWN use the counterfactual policy or stick to the current one?

Open Bandit Pipeline: Simple OPE Implementation

We can easily implement OPE itself and experiments on OPE

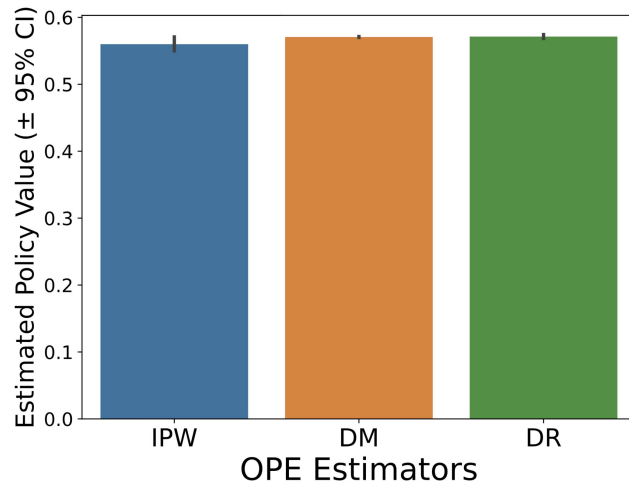
```
# a case for implementing OPE of the BernoulliTS policy using log data generated by the Random policy
from obp.dataset import OpenBanditDataset
from obp.policy import BernoulliTS
from obp.simulator import run_bandit_simulation
from obp.ope import OffPolicyEvaluation, ReplayMethod

# (1) Data loading and preprocessing
dataset = OpenBanditDataset(behavior_policy='random', campaign='women')
bandit_feedback = dataset.obtain_batch_bandit_feedback()

# (2) Offline Bandit Simulation
counterfactual_policy = BernoulliTS(n_actions=dataset.n_actions, len_list=dataset.len_list)
selected_actions = run_bandit_simulation(bandit_feedback=bandit_feedback, policy=counterfactual_policy)

# (3) Off-Policy Evaluation
ope = OffPolicyEvaluation(bandit_feedback=bandit_feedback, ope_estimators=[ReplayMethod()])
estimated_policy_value = ope.estimate_policy_values(selected_actions=selected_actions)

# estimated performance of BernoulliTS relative to the ground-truth performance of Random
relative_policy_value_of_bernoulli_ts = estimated_policy_value['rm'] / bandit_feedback['reward'].mean()
print(relative_policy_value_of_bernoulli_ts) # 1.120574...
```



Democratizing and Standardizing Off-Policy Evaluation

Open Bandit Pipeline: Documentation

We built a detailed [documentation](#) of the Open Bandit Pipeline

 obp
latest

INTRODUCTION:
[About](#)
[Overview of Off-Policy Evaluation](#)

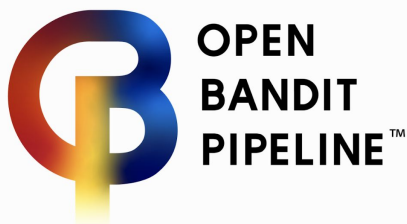
GETTING STARTED:
[Installation](#)
[Quickstart](#)

PACKAGE REFERENCE:
[OBP Package Reference](#)

OTHERS:
[Github](#)
[References](#)


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Docs » Open Bandit Pipeline; a python library for bandit algorithms and off-policy evaluation
[Edit on GitHub](#)



Open Bandit Pipeline; a python library for bandit algorithms and off-policy evaluation

Overview

Open Bandit Pipeline (OBP) is an open source end-to-end python library for bandit algorithms and off-policy evaluation (OPE). The toolkit comes with the *Open Bandit Dataset*, a large-scale logged bandit feedback data collected on a fashion e-commerce platform, [ZOZOTOWN](#). The purpose of the open data and library is to enable easy, realistic, and reproducible evaluation of bandit algorithms and OPE. OBP has a series of implementations of dataset preprocessing, bandit policy interfaces, offline bandit simulator, and standard OPE estimators.

Our open data and pipeline facilitate evaluation and comparison related to the following research topics.

Table of Contents (ToC)

- Overview of OPE
- Evaluation of OPE
- Dataset Description
- Package References
- Related Work

Real-World Applications:

Estimator Selection and
Counterfactual Policy Search

Application1: Off-Policy Estimator Selection

RQ: Which is the best estimator in DM/IPW/SNIPW/DR?

1. Apply the evaluation of OPE protocol on the platform by the software

Table 4: Comparing Relative-Estimation Errors of OPE Estimators (**Random** → **Bernoulli TS**)

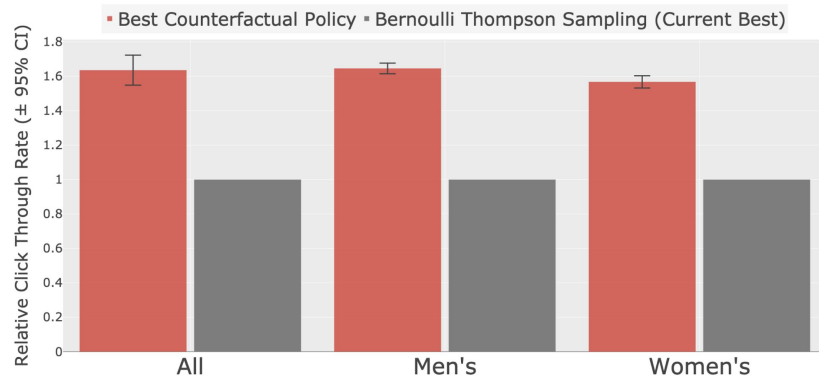
Estimators	Campaigns		
	All	Men's	Women's
DM	0.23879 [0.22998, 0.24988]	0.24155 [0.22656, 0.25592]	0.22884 [0.22224, 0.23423]
IPW	0.03477 [0.01147, 0.06592]	0.09806 [0.07485, 0.12151]	0.03252 [0.01708, 0.04912]
SNIPW	0.03381 [0.01005, 0.06662]	0.08153 [0.05677, 0.10592]	0.03179 [0.01562, 0.04825]
DR	0.03487 [0.01094, 0.06784]	0.08528 [0.06186, 0.10876]	0.03224 [0.01605, 0.04843]

Result: SNIPW is the most accurate estimator on the ZOZOTOWN platform

Application2: Counterfactual Policy Search

RQ: Can we find the promising counterfactual policy by OPE?

1. Construct a set of candidate counterfactual policies
2. Apply the most accurate estimator (SNIPW) to the candidate policies
3. Compare the counterfactual policy selected by OPE and the current one



Result:

We can identify the counterfactual policy that improves the CTR about 40-60%

Figure 3: Comparing Counterfactual Policy and Bernoulli Thompson Sampling

Summary

- Accurate OPE enables the **safe policy improvements in theory**
- Experiments in OPE papers are *unrealistic* or *irreproducible*
 - which creates the critical gap in theory and practice
- We release the *open-source dataset/software*
 - enable the *realistic* and *reproducible* experiments on OPE
 - make it easier to use OPE in practice
 - their benefits are verified on our fashion EC platform

Thank you!

email: saito@hanjuku-kaso.com

paper: <https://arxiv.org/abs/2008.07146>

github: <https://github.com/st-tech/zr-obp>

docs: <https://zr-obp.readthedocs.io/en/latest/>

google group: <https://groups.google.com/g/open-bandit-project>

dataset: <https://research.zozo.com/data.html>